

# Procta

## AI-Proctored Exam Platform

Feature inventory · as of 30 May 2026

What's actually shipped — sourced from the codebase, not the roadmap. Every entry below corresponds to working code in production today.

## 1. Live Proctoring Engine

Multi-signal AI runs in the locked Electron browser on the student's machine. Frames stream to the teacher dashboard in real time over WebSocket; events stream over Server-Sent Events.

|                                                   |                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Face presence</b>                              | Detects when the student leaves the camera. InsightFace embedding model (CPU). Logged as <i>face_missing</i> violation.                                                                                                                                                                         |
| <b>Wrong-person detection</b>                     | Embedding compared against the ID-verification selfie at calibration. Triggers if a different person sits at the keyboard mid-exam. <i>wrong_person</i> violation, weighted heaviest in risk score.                                                                                             |
| <b>Multi-face detection</b>                       | Two faces in frame for >2 s → <i>multiple_faces</i> violation. Catches another person looking over the shoulder.                                                                                                                                                                                |
| <b>Gaze tracking</b>                              | MediaPipe face-mesh + 9-point calibration grid. Logs sustained off-screen gaze. Per-student calibration quality tier (Tight / Loose / Suspicious) shown in Results.                                                                                                                             |
| <b>Head-pose estimation</b>                       | Yaw / pitch / roll from face landmarks; gating threshold tuned in calibration to per-student baseline (hardware-aware).                                                                                                                                                                         |
| <b>Cheat-object detection (YOLOv8 + SAHI)</b>     | Phone, headphones, earbuds (custom ear-crop classifier). Slicing-aided inference for objects partially out of frame. GPU auto-detect: CUDA → MPS → CPU.                                                                                                                                         |
| <b>Audio anomaly (RMS)</b>                        | Sustained voice / conversation-pattern detection via sounddevice RMS. Tuned to ignore typing and ambient noise. Three event types: <i>voice_detected</i> , <i>sustained_voice</i> , <i>conversation_detected</i> .                                                                              |
| <b>On-device speech-to-text keyword detection</b> | Vosk (en-IN + hi-IN) runs on the student's CPU and matches transcripts against a built-in cheat-phrase list plus per-exam teacher-added keywords. Fires <i>keyword_uttered</i> on match with the matched phrase and a ~5 s transcript snippet. Audio never leaves the device.                   |
| <b>Multi-voice detection</b>                      | Silero VAD gates voice-active segments; python_speech_features MFCC vectors clustered over a rolling 60 s window via a numpy-only 2-cluster silhouette check. Two distinct voices in the buffer fires <i>multiple_voices_detected</i> . Catches an off-camera helper that RMS alone would miss. |
| <b>Behavioural correlation engine</b>             | Multi-signal fusion: gaze + audio + face-pose patterns scored together to surface coordinated cheating (e.g. consistent gaze drift right + faint voice spikes).                                                                                                                                 |

## 2. Lockdown Browser (Electron)

The student's exam runs in a custom Electron build. The OS session is locked while the exam is active.

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| <b>Tab-switch prevention</b> | Browser focus loss for >1 s → <i>tab_switch</i> violation. |
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| <b>Process integrity scan</b>           | Live scan for VPN, remote-desktop, screen-share, debugger, AI assistant tools. Hyphen-aware regex (e.g. parsec-cloud no longer false-positives parsec). |
| <b>Hardware-locked session</b>          | Camera + microphone permissions granted at calibration; lost permission mid-exam ends the session.                                                      |
| <b>Auto-update via electron-updater</b> | Hits GitHub Releases on launch. Existing installs upgrade on next relaunch with no manual intervention.                                                 |
| <b>Offline answer resilience</b>        | LocalStorage backup of every keystroke. On reconnect, background bulk-save catches up. Students never lose answers to a network blip.                   |
| <b>Mac code-signed</b>                  | Notarised .dmg distributed via /download/mac and /download/mac-x64. Windows code-signing pending Azure Trusted Signing.                                 |

### 3. Teacher Dashboard

Real-time control plane at app.procta.net. Single-page app with sticky chrome and single-scroll content per pattern.

|                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Live sessions</b>                                           | 47-student grid view with live face-frames over WebSocket, auto-updating risk scores, severity badges, force-submit, broadcast messages.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>AI live-risk triage</b>                                     | One-line LLM summary per session, cached 60 s in Redis. Surfaces the 3 students most likely to be cheating right now.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Forensics timeline</b>                                      | Per-session scrub-through of every violation event with linked evidence screenshots. Downloadable PDF report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Results table</b>                                           | Sortable, filterable, with per-row risk scores, calibration tier, and CSV / Excel / PDF / scorecard-ZIP exports.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Post-exam analytics</b>                                     | Score distribution (10 buckets), per-question difficulty + discrimination index, violation heatmap, risk band breakdown. Cached 60 s.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Real-time risk alerts</b>                                   | SSE toast notifications when a student's risk score crosses thresholds, with deep-link to the timeline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Question editor + lint</b>                                  | MCQ / multi-select / true-false / short-answer types. AI lint runs each question pre-publish to surface ambiguity, leakage, or factual errors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Question bank + import / export</b>                         | Reusable pool of questions across exams. CSV / JSON bulk import with preview. Tag-based filtering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Exam templates</b>                                          | Save an exam configuration + question set as a template; clone for next semester or section in one click.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Student groups / sections</b>                               | Restrict exam access by group membership. Bulk add from registered students.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Live chat with students</b>                                 | Bidirectional WebSocket chat. Teacher broadcast or 1:1. Chat hub bounded with eviction (50 sockets/tenant, 4 h metadata TTL).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Live teacher intervention (Warn / Pause / Resume / End)</b> | Three escalating actions on each live-session row. Warn pushes an amber banner + chime to the student with a chip-coded reason (eyes_off_screen / phone_visible / talking_to_someone / multiple_tabs / other). Pause locks the student UI with a full-screen overlay AND stops the exam timer; resume credits the paused interval back. End requires re-authentication and a chip-coded reason (academic_dishonesty / identity_fraud / environment_issue / repeated_violations / student_request / technical_failure / other); the reason persists on the session row, embeds in the audit-trail violation, and surfaces in the scorecard PDF + CSV export. |

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| <b>Cluster review (bulk false-positive triage)</b> | Groups not-yet-dismissed violations by (exam, type, severity) across an entire cohort. One-click bulk dismiss with a reason stamps <code>dismissed_at</code> + <code>dismissed_reason</code> on every row in scope. Critical at 3,500-student exam scale where per-session review doesn't fit a teacher's afternoon. |
| <b>Onboarding wizard</b>                           | 5-step intro for first-time teachers. Persists <code>procta_onboarded</code> in <code>localStorage</code> ; ? button re-opens.                                                                                                                                                                                       |

## 4. Student Experience

Email invite → desktop install → calibration → exam. Calibration is mandatory and quality-scored.

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| <b>Invite-driven enrolment</b>          | Teacher sends email invite; clicking the link auto-creates the student account on first <code>/validate-student</code> call. No pre-registration needed. |
| <b>9-point gaze calibration</b>         | Mandatory before exam start. Quality tier baked into the session record so teachers can flag suspicious-tight calibrations later.                        |
| <b>Continuous identity verification</b> | Face embedding sampled throughout calibration; mismatch aborts the exam with a typed <code>calibration_abort</code> event that pages the teacher.        |
| <b>ID-document verification</b>         | Selfie + photo-ID upload. Stored as evidence; teacher reviews in dashboard pending-verifications panel.                                                  |
| <b>In-exam chat with invigilator</b>    | Floating action button on the exam window. Encrypted, auth-scoped to the session.                                                                        |
| <b>Practice mode</b>                    | Sandbox <code>session_id</code> ( <code>PRACTICE_...</code> ) that exercises the full flow against canned questions. No DB writes.                       |
| <b>Multi-exam lobby</b>                 | Single auth → list of available exams → pick one → calibrate → write.                                                                                    |
| <b>Student performance history</b>      | Longitudinal view across all exams the student has taken with this teacher (when student account is connected).                                          |

## 5. Grading and Assessment

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| <b>Auto-graded MCQ / multi-select / T-F</b>   | Scored at submit. Multi-select uses set-equality. Shuffled-option mappings preserved per-session for accurate grading.                                                                                                                                                                                                       |
| <b>Short-answer with AI-suggested grading</b> | Teacher writes reference + rubric + max_score. LLM proposes score + 1-2 sentence rationale + confidence (high/med/low). Teacher confirms or overrides; only <i>teacher_score</i> counts in the gradebook.                                                                                                                    |
| <b>Rolled-up scoring</b>                      | On grade-confirm, MCQ correct + sum(teacher_score) auto-recomputes the session total + percentage. Idempotent — safe to re-run.                                                                                                                                                                                              |
| <b>Per-student scorecard PDF</b>              | Header, score summary, per-question table (student answer vs reference, correct/incorrect), violation summary, calibration tier. When the session was force-submitted by a teacher, a Termination row surfaces the reason chip + the free-text note so the document is defensible in a later grade-challenge or DPDP review. |
| <b>Bulk scorecard ZIP</b>                     | Streams scorecards for an entire exam as a ZIP via BytesIO. Doesn't load all PDFs into memory.                                                                                                                                                                                                                               |
| <b>Email scorecards</b>                       | Idempotent per <i>scorecard_emailed_at</i> . Teacher can click twice; already-emailed students skipped.                                                                                                                                                                                                                      |
| <b>Composite risk score (0–100)</b>           | Log-saturating per-violation-type weights, duration-normalised. Cached. Recomputable via admin backfill endpoint.                                                                                                                                                                                                            |
| <b>AI session narrative</b>                   | 1-paragraph human-readable summary of a student's session (LLM over the last 80 violation events). Cached 60 s.                                                                                                                                                                                                              |

## 6. Operations and Infrastructure

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>FastAPI + async stack</b>                            | All hot-path endpoints async via <i>_atable</i> ; supabase client wraps run in a worker thread for sync calls. 300+ concurrent students target on \$6 droplet.                                                                                                                                                                                                                                                                                                                    |
| <b>Hardware governor (CPU-adaptive proctor cadence)</b> | Reads CPU% via psutil every 5 s on the student machine. When CPU > 85% for two consecutive samples, ML inference drops to 1 frame per 2 s (event-only mode) and the audio worker skips every other ASR pass; rampback when CPU < 60%. Keeps the exam UI responsive on Rs. 30,000 budget Lenovo IdeaPad-class laptops instead of freezing under thermal throttling. Logs <i>client_throttled</i> (info severity) so teachers can correlate 'student exam felt slow' with hardware. |

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| <b>On-device model download bootstrap</b> | scripts/download_audio_models.py invoked from Electron after pip install completes. Pulls Vosk en-IN + hi-IN + Silero VAD into ./weights/ with SHA-256 verification. Idempotent; cached after first run. Falls through to the existing RMS-only voice path if a download fails so the proctor never blocks on it. |
| <b>Router decomposition</b>               | 8 domain routers: auth / exam / admin / question_bank / grading / public / sse / chat. Single 7,000-line main.py shrunk to 382.                                                                                                                                                                                   |
| <b>Pydantic strict-mode validation</b>    | ConfigDict(strict=True) on every request body model. Rejects type-coercion attacks (e.g. "123" for an int field) at the boundary.                                                                                                                                                                                 |
| <b>Redis + in-memory caching</b>          | Two-layer: per-process LRU + Redis. Sorted-set LRU eviction for live frames (50-session cap). Cache-keyed risk scores, analytics, AI triage.                                                                                                                                                                      |
| <b>Atomic invite cap</b>                  | Postgres RPC claim_invite_cap uses conditional UPDATE under row lock. Concurrent senders can't overshoot the daily cap.                                                                                                                                                                                           |
| <b>Streaming CSV / PDF exports</b>        | StreamingResponse + async generators. Doesn't OOM on 5,000-row results tables.                                                                                                                                                                                                                                    |
| <b>Structured JSON logs + request IDs</b> | Every request gets an X-Request-ID; logs are JSON for grep + log-aggregation tooling.                                                                                                                                                                                                                             |
| <b>Sentry integration (optional)</b>      | SENTRY_DSN env var enables capture; off by default for local dev.                                                                                                                                                                                                                                                 |
| <b>Security headers</b>                   | CSP, HSTS, X-Frame-Options DENY, Permissions-Policy (camera + mic blocked at the app level for non-exam paths).                                                                                                                                                                                                   |
| <b>ETag + conditional requests</b>        | Middleware adds ETags to JSON responses $\leq 1$ MB; 304 on matching If-None-Match.                                                                                                                                                                                                                               |
| <b>GZip response compression</b>          | 60-80% reduction on JSON >500 B. Static assets pre-gzipped at Docker build, served by Caddy.                                                                                                                                                                                                                      |
| <b>Auto-update pipeline</b>               | GitHub Releases auto-discovered by backend (10 min TTL). Download buttons resolve to the latest release without redeploy.                                                                                                                                                                                         |
| <b>Migration discipline</b>               | 15 numbered SQL migrations, all idempotent. PGRST204 schema-cache drift handled with runtime fallback.                                                                                                                                                                                                            |
| <b>Test suite</b>                         | 14 pytest files: alerts, sessions, history, summary, behavioural analysis, e2e proctor, forensics timeline, invites, supporting modules. CI excludes heavy proctor tests via marker.                                                                                                                              |

## 7. Communication and Workflow

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| <b>Email invites with reminders</b>   | 1 h + 24 h reminder cadence for unaccepted invites. Tracked via <code>opened_at</code> , <code>clicked_at</code> , <code>accepted_at</code> . |
| <b>Real-time teacher↔student chat</b> | Reconnection-tolerant WS. Per-session message history bounded by deque + cleanup loop.                                                        |
| <b>Broadcast announcements</b>        | Teacher → all live students in an exam. Persists in chat history.                                                                             |
| <b>Force-submit + clear session</b>   | Time-windowed clear tokens (60 s TTL); audit-trailed violation event for every clear.                                                         |
| <b>Webhook outbound (planned)</b>     | Schools' SIS systems. Stub exists; payload schema pending.                                                                                    |

## 8. Privacy and Compliance Posture

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| <b>RLS policies on all tables</b>     | Every table enforces <code>teacher_id</code> filtering at the DB level via Supabase RLS. Even if the API has a bug, cross-tenant data is unreachable.                                                                                                                                                                                |
| <b>JWT-scoped session ownership</b>   | Session keys embed <code>teacher_id</code> UUID; access validated against JWT <code>tid</code> claim on every authenticated call.                                                                                                                                                                                                    |
| <b>Encrypted at rest</b>              | Supabase Postgres encryption + DigitalOcean volume encryption.                                                                                                                                                                                                                                                                       |
| <b>HTTPS-only + HSTS</b>              | Caddy auto-provisions Let's Encrypt; HSTS preload-eligible.                                                                                                                                                                                                                                                                          |
| <b>Zero raw-video storage</b>         | Camera frames are processed on the student's device by the on-device ML stack. Only JPEG snapshots of flagged moments are uploaded as evidence; no continuous video reaches Procta servers. Materially reduces both the DPDP-aligned consent surface and the operating cost of storage.                                              |
| <b>Zero raw-audio storage</b>         | Microphone capture is analysed by the on-device Vosk + Silero VAD + MFCC pipeline. Only event metadata ( <code>keyword uttered</code> with the matched phrase, <code>multiple_voices_detected</code> with a confidence score) and the synchronously-captured camera JPEG are uploaded. Raw audio never leaves the student's machine. |
| <b>Auto-delete evidence retention</b> | Cron in <code>entrypoint.sh</code> auto-purges evidence JPEGs older than <code>SCREENSHOT_RETENTION_DAYS</code> (default 90). Aligned with DPDP storage limitation and the common 90-day exam-record retention norm in Indian higher education.                                                                                      |

## 9. Roadmap Snapshot

Items with working code or migrations queued. Prioritised by school-procurement leverage.



|                                      |                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Windows code signing</b>          | Azure Trusted Signing \$10/mo. Removes SmartScreen warning for Windows downloads — currently deal-blocker for school IT. |
| <b>Mobile PWA (lite proctoring)</b>  | Camera-based proctoring still works; tab-switch prevention doesn't. Position as low-stakes-quiz tool.                    |
| <b>Inbuilt coding questions</b>      | Pyodide for Python in browser, LLM-judged for C++/Java. Phase 13a designed; build deferred per founder.                  |
| <b>Per-criterion rubric grading</b>  | Extend short-answer to (clarity / correctness / depth) × marks. Universities specifically asked.                         |
| <b>Cross-cohort plagiarism check</b> | Cosine sim on embeddings of every short-answer pair. Cheap with a small embedding model.                                 |
| <b>Two-factor for teacher login</b>  | TOTP via authenticator app.                                                                                              |
| <b>Bulk exam scheduling</b>          | Clone exam to N sections × M time slots. Coaching-institute ergonomic win.                                               |

## 10. Future Vision

Bigger bets — 6–24 months out. Not on the immediate sprint, but the direction Procta evolves to become a category-defining platform rather than a proctoring tool.

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>React dashboard rewrite</b>                     | The current 7,000-line vanilla-JS dashboard is functional but ages every time a feature lands. Migrate to a React + Vite SPA matching the marketing site's design system. Wins: component reuse, dark/light theme switching, keyboard shortcuts, fluid loading states, drag-to-reorder, auto-escaping (kills the entire XSS class), and a real design language for sales screenshots.                        |
| <b>Inbuilt coding questions</b>                    | Pyodide for Python (browser-side WASM, zero server load). LLM-judged for C++/Java/Go where running untrusted code isn't worth the infra. Monaco editor with syntax highlight + autocomplete. Tests run on Run, hidden tests run on Submit. Phase-13a designed in Plan; deferred per founder.                                                                                                                 |
| <b>AI exam generator from syllabus</b>             | Upload a PDF syllabus or paste a topic list; LLM produces a balanced 40-question exam tagged by Bloom's level + difficulty. Teacher tweaks → publish. Saves the 2-hour exam authoring session that's currently every teacher's worst job.                                                                                                                                                                    |
| <b>On-device speaker identification (deferred)</b> | Today's multi-voice detection answers "two voices yes/no" via MFCC clustering. Speaker identification would add a stable label per voice ("voice A" / "voice B") and track when each appears. Requires either a pyannote ONNX export (~30 MB additional model) or a custom siamese-network fingerprint head. Worth doing only if customer feedback shows the yes/no signal is leaving evidence on the table. |
| <b>Native mobile apps (iOS + Android)</b>          | React Native shell wrapping the same proctoring engine compiled for ARM. Position as the primary tool for low/medium-stakes assessments where the desktop install is friction. Camera-based proctoring still works; lockdown is weaker but compensated with stricter face-presence policy.                                                                                                                   |
| <b>Teacher mobile app</b>                          | Push-notify on high-risk events. Swipe-right to acknowledge, swipe-left to force-submit. For invigilators monitoring remotely. Pairs with the desktop dashboard, doesn't replace.                                                                                                                                                                                                                            |
| <b>LMS integrations</b>                            | Canvas, Moodle, Blackboard, Google Classroom plugins. Single-sign-on, exam launch from inside the LMS, scores auto-pushed back. The integration story is what unblocks university procurement deals.                                                                                                                                                                                                         |
| <b>Adaptive testing (IRT-based)</b>                | Questions adjust difficulty to the student's demonstrated level mid-exam, using item-response theory. Same total time, more accurate measurement, harder to brute-force-cheat (no two students see the same question sequence).                                                                                                                                                                              |

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| <b>Teacher AI assistant</b>               | Chat with your gradebook in natural language: <i>“Which students improved most this semester?” “Show me question #14’s discrimination index across all sections.”</i> Backed by an LLM with read-only tool access to the teacher’s data. |
| <b>AI proctor explainer for students</b>  | When a student is flagged, they see a plain-English explanation: <i>“Your gaze drifted off-screen for 12 s during Q14, which is longer than 95% of test-takers.”</i> Removes the “proctoring is a black box” hostility.                  |
| <b>Cross-cohort plagiarism check</b>      | Cosine similarity on embeddings of every short-answer pair across the cohort. Highlights suspicious clusters. Cheap with a small embedding model run nightly.                                                                            |
| <b>Live invigilator marketplace</b>       | When AI flags a high-risk session, optionally route to a human reviewer (vetted, on demand, paid per minute). Schools that don’t have invigilator capacity buy review time from the platform. Two-sided revenue stream.                  |
| <b>Multi-monitor detection</b>            | Browser screen-API + webcam reflection analysis to detect secondary monitors. Currently the loophole every prep-school student exploits.                                                                                                 |
| <b>rPPG attention + stress metrics</b>    | Heart-rate variability inferred from sub-pixel facial colour changes (research-grade rPPG). Surface attention dropoff and stress to teachers without storing biometric data — computed live, never persisted.                            |
| <b>Whiteboard / stylus mode</b>           | Students show working on iPad / Wacom tablet for math + physics exams. Stroke-by-stroke recording for grading and review. OCR + LLM for partial credit on handwritten derivations.                                                       |
| <b>Federated SSO for school districts</b> | SAML / OIDC integration so a district admin manages 50 schools’ teachers from one console. Critical for K-12 district sales.                                                                                                             |
| <b>Public API + developer SDK</b>         | Third-party integrations on top of Procta — e.g. coaching institutes plugging Procta into their custom LMS. Rate-limited, scoped tokens, OpenAPI spec. Becomes a moat.                                                                   |
| <b>On-device LLM for free tier</b>        | Llama / Phi running locally for grading, lint, and risk narratives. Removes Groq dependency and cuts inference cost to zero for paying schools. Slower but private — useful selling point in EU.                                         |
| <b>Exam marketplace</b>                   | Teachers publish question banks; others subscribe. Teachers earn revenue on usage. 30/70 split, Procta keeps 30%. Compounds the network effect once question-bank UX matures.                                                            |

*Generated from the Procta codebase. Sections 1–8 list features with shipping code or tracked migrations. Section 9 is near-term roadmap (working code or designs queued). Section 10 is directional — bigger bets that define where Procta goes next.*